

Homework 6

Release Date: October 29, 2025

Due Date: November 5, 2025

Submission Instructions: Submit a single PDF (clear scans or photos compiled) to Gradescope and assign pages for each problem. Show key steps and justify answers.

Collaboration & AI Policy: You may discuss approaches with classmates, but write up your own solutions and list collaborators. If you use computational tools (including LLMs) for checking, cite them and ensure the reasoning is your own.

Problem 1: The Gram-Schmidt Process (10 points)

Let V be an inner product space with a basis $\mathcal{B} = (b_1, \dots, b_n)$. Our goal is to convert this into an *orthonormal basis* $\mathcal{U} = (u_1, \dots, u_n)$, where all vectors are mutually orthogonal and have unit norm.

The Algorithm. The Gram-Schmidt process accomplishes this iteratively.

Input: A basis $(b_1, \dots, b_n)$
Output: An orthonormal basis $(u_1, \dots, u_n)$

// Initialize with the first vector;
 $v_1 \leftarrow b_1$;
 $u_1 \leftarrow \frac{v_1}{\|v_1\|}$;

// Iterate and orthogonalize the rest
for $k = 2, \dots, n$ **do**
 $v_k \leftarrow b_k - \sum_{i=1}^{k-1} \langle b_k, u_i \rangle u_i$;
 $u_k \leftarrow \frac{v_k}{\|v_k\|}$;
end

Proof of Correctness. In this problem, you will prove that this process works. The core idea is to show by induction that for any $k \in \{1, \dots, n\}$, the set $(u_1, \dots, u_k)$ is an orthonormal basis for the subspace $\text{span}(b_1, \dots, b_k)$.

- Base Case (k=1):** (2 points) Following the algorithm, the first step is $v_1 = b_1$ and $u_1 = \frac{b_1}{\|b_1\|}$. Show that $\{u_1\}$ is an orthonormal set and that $\text{span}(u_1) = \text{span}(b_1)$.
- Inductive Step (Orthogonality):** (4 points) Assume that for some $k < n$, the set $(u_1, \dots, u_k)$ is orthonormal. Using the definition of v_{k+1} from the algorithm, show that it is orthogonal to u_j for all $j \in \{1, \dots, k\}$. This proves that $(u_1, \dots, u_{k+1})$ is an orthonormal set.
- Inductive Step (Span):** (4 points) Assume that $\text{span}(u_1, \dots, u_k) = \text{span}(b_1, \dots, b_k)$. Show that the span is preserved at the next step, i.e., $\text{span}(u_1, \dots, u_{k+1}) = \text{span}(b_1, \dots, b_{k+1})$.

By induction, this process proves that $(u_1, \dots, u_n)$ is an orthonormal basis for V .

Problem 2: Representer Theorem (15 points)

In lecture, we proved the Representer Theorem for a regularizer of the form $\lambda\|f\|_{\mathcal{H}_K}^2$, which penalizes solutions with large norms and effectively shrinks the optimal function f^* towards the zero function.

In this exercise, we will prove a more general version where the regularizer incorporates a **prior** or **reference function** $h \in \mathcal{H}_K$. This function h represents a baseline belief about the solution, and the new regularizer, $\lambda\|f - h\|_{\mathcal{H}_K}^2$, penalizes functions that deviate from this prior.

The goal is to solve the following optimization problem:

$$f^* = \arg \min_{f \in \mathcal{H}_K} \left[\sum_{i=1}^N \ell(f(x_i), y_i) + \lambda\|f - h\|_{\mathcal{H}_K}^2 \right]$$

where $h \in \mathcal{H}_K$ is a fixed function.

Let $U = \text{span}\{K_{x_1}, \dots, K_{x_N}\}$. Then we can project both f and h onto this subspace and its orthogonal complement where $f = f_U + f_{U^\perp}$ and $h = h_U + h_{U^\perp}$.

Your task is to prove the theorem by following these steps, which closely mirror the proof from the lecture.

Step 1: Decompose the regularizer (4 points). Show that the norm decomposes along the orthogonal subspaces, i.e.,

$$\|f - h\|_{\mathcal{H}_K}^2 = \|f_U - h_U\|_{\mathcal{H}_K}^2 + \|f_{U^\perp} - h_{U^\perp}\|_{\mathcal{H}_K}^2 \quad (1)$$

Step 2: Analyze the loss term (2 points). Using the reproducing property of $\mathcal{H}_K$, show that the predictions at the data points only depend on the projection onto U :

$$f(x_i) = f_U(x_i) \quad (2)$$

Conclude that the loss term $\sum_{i=1}^N \ell(f(x_i), y_i)$ is independent of the orthogonal component $f_{U^\perp}$.

Step 3: Combine and conclude (6 points). Use the results from the previous steps to argue that the optimal solution f^* must satisfy $f_{U^\perp}^* = h_{U^\perp}$. Then, show that this implies the solution must have the form:

$$f^* = \sum_{i=1}^N \alpha_i K_{x_i} + h \quad (3)$$

for some coefficients $\alpha_i \in \mathbb{R}$.

Step 4: A special case (3 points). Show that if the reference function h lies completely within the subspace U (i.e., $h \in U$), the result in (3) can be simplified to:

$$f^* = \sum_{i=1}^N \beta_i K_{x_i} \quad (4)$$

for some (potentially different) coefficients $\beta_i \in \mathbb{R}$.